Jason Lee

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Education

The University of Texas at Austin — Bachelor of Science, Computational Engineering Honors — *May 2027*
Minor: Robotics | GPA: 3.7/4.0

Global Education - Kyoto, Japan — Completed major-related coursework abroad — *June 2024 - August 2024*

Skills and Interests

Languages: Python, C++, C#, MATLAB, Java, JavaScript/TypeScript, LaTeX
Frameworks & Technologies: FastAPI, Flask, REST API, TensorFlow, PyTorch, CUDA, pandas, scikit-learn, NumPy, React
Developer Tools: Git, Linux, Docker, Kubernetes, Redis, Jupyter, GitHub, GitLab, Node.js, ROS, AWS, Confluence, Jira
Interests: Production ML, Platform Engineering, Deep Learning, Machine Learning, Marathons, Chess, Bouldering

Work Experience

IBM — **Austin, TX**
Agentic AI Intern — *May 2026 - Present*
- Engineered a least-privilege permission inference engine for LLM tool-calling agents using the MiniScope algorithm, dynamically enforcing fine-grained, context-aware authorization across IBM Verify's agentic AI orchestration pipeline
- Implemented OAuth 2.0 dynamic token exchange to enforce minimal scopes across multi-agent agentic AI workflows
- Built A2A proof-of-concept for agentic tool invocation validating permission inference accuracy and sub-second latency

Samsung — **Austin, TX**
Software Engineering Intern — *May 2025 - August 2025*
- Refactored app publish script in C# for cloud infrastructure platform used by **80%** of employees and **2,400** apps monthly
- Enhanced internal app publish speed by **315%** and reduced first-time publishing mistakes by **20%** by streamlining UI
- Automated Kubernetes orchestration with KubeAuto API, boosting CI/CD efficiency and deployment reliability
- Utilized Agile methodologies, GitLab code reviews, and Jira for sprint planning in the software deployment process

Cross Labs — **Kyoto, Japan**
AI Research Intern — *June 2024 - October 2024*
- Engineered a neuroevolution system with CMA-ES and PyTorch to train a 7-DOF Panda robotic arm for object manipulation in MuJoCo/robosuite simulation, achieving **3.3**× reward improvement across **50 generations**
- Integrated CUDA acceleration into neural network training pipelines, reducing training time by over **3000%**
- Developed a deep neural network Jupyter Notebook tutorial for beginners and future interns new to machine learning

Projects

StrumSense | *Audio ML, Deep Learning, Python, React/Next.js* — *November 2025 - December 2025*
- Built a full-stack, multi-stage ML pipeline using OpenL3 CNN embeddings, Librosa DSP features, and weighted cosine similarity to generate personalized song recommendations from a **1000** track dataset based on the user's input
- Implemented async inference with Python workers, caching, and parallelized Last.fm API enrichment, reducing model inference by **60%** to achieve **sub-3**-minute end-to-end latency with Dockerized microservices on Render and Vercel

Echoes of the Unknown | *Python, Flask, Docker, Kubernetes* — *January 2025 - May 2025*
- Constructed a containerized REST API analyzing **80,000** UFO sightings using Flask, Redis, HotQueue, and Kubernetes
- Designed a microservices system with persistent storage and scalable workers for asynchronous job processing

Twilight | *C++, Unreal Engine* — *August 2023 - May 2024*
- Developed a 2D in 3D action-adventure game using C++ and Unreal Engine 5 Blueprints with the PaperZD plugin
- Implemented AI to optimize procedural animation, achieving **20%** improvement in gameplay dynamics and FPS

Leadership Experience

Amazon - *Amazon Robotics Mentee* — *September 2025 - Present*
- Accepted into Amazon Guides, a selective **8**-month professional development program for high-achieving talent
- Schedule meetings with my Amazon Robotics mentor to gain insight and define goals that will guide my career growth

Student Engineering Council - *Committee Member* — *August 2024 - Present*
- Organize **15+** large-scale engineering events, manage **$20,000** in budget, and engage **70%** of UT engineering students
- Mentor **20+** freshmen and host opportunities that connect over **6,200** freshmen with companies and local organizations